Namespace BOOSEapp

Classes

AdditionalFeatures

Provides additional BOOSE features such as random values, user input, and named colours.

AppCanvas

Custom canvas implementation used by the BOOSE interpreter. Responsible for drawing shapes and managing the drawing cursor.

AppCommandFactory

Custom command factory used to create BOOSE commands. It extends the default command factory and returns application-specific command implementations.

ApplicationConfiguration

Bootstrap the application configuration.

Array

Array interpreter extension for BOOSE, adding support for array variables. Supports int variables, real variables, int arrays, real arrays, together with poke, peek and simple write commands.

Circle

BOOSE command that draws a circle on the canvas.

Clear

Clears the drawing canvas.

Form1

Main user interface for the BOOSE interpreter. Allows the user to input BOOSE programs, execute them, and display graphical output on the canvas.

If

Handles IF, ELSE and END IF logic in BOOSE programs.

Int

Simple interpreter extension that adds support for integer variables (int). Allows declaration, assignment and usage of variables in BOOSE commands.

LogInDebug

Singleton logger used to write debug information.

Maths

Provides simple mathematical expression evaluation (+, -, *, /)

Method
: Simple interpreter that supports integer methods in BOOSE. Methods do not return values explicitly – instead, the result is stored in a variable with the same name as the method.

MoveTo
: BOOSE command that moves the drawing cursor to a specified position on the canvas.

Program

Real
: Real variable support for BOOSE interpreter. Allows declaration, assignment and usage of variables in BOOSE commands.

Rectangle
: BOOSE command that draws a rectangle on the canvas.

Tri
: BOOSE command that draws a triangle on the canvas.

Interfaces

IInterpreter
: Common interface for classes that can run a BOOSE program.

Class AdditionalFeatures

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Provides additional BOOSE features such as random values, user input, and named colours.

```
public class AdditionalFeatures
```

Inheritance

object ← AdditionalFeatures

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AdditionalFeatures(AppCanvas, Dictionary<string, int>)

Initialises the additional features helper.

```
public AdditionalFeatures(AppCanvas canvas, Dictionary<string, int> intVars)
```

Parameters

`canvas` AppCanvas

Canvas used for drawing commands.

`intVars` Dictionary<string, int>

Integer variables used by the interpreter.

Fields

canvas

```
private AppCanvas canvas
```

Field Value

AppCanvas

intVars

```
private Dictionary<string, int> intVars
```

Field Value

Dictionary<string, int>

rnd

```
private Random rnd
```

Field Value

Random

Methods

TryExecute(string)

Tries to execute an additional feature command.

```
public bool TryExecute(string line)
```

Parameters

`line` string

Single line of BOOSE code.

Returns

bool

True if the command was handled; otherwise false.

Class AppCanvas

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Custom canvas implementation used by the BOOSE interpreter. Responsible for drawing shapes and managing the drawing cursor.

```
public class AppCanvas : ICanvas
```

Inheritance

object ← AppCanvas

Implements

ICanvas

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AppCanvas(int, int)

Initializes a new drawing canvas with the specified size.

```
public AppCanvas(int xsize, int ysize)
```

Parameters

`xsize` int

Width of the canvas.

`ysize` int

Height of the canvas.

Fields

CanvasBitmap

```
private Bitmap CanvasBitmap
```

Field Value

Bitmap

Pen

```
private Pen Pen
```

Field Value

Pen

g

```
private Graphics g
```

Field Value

Graphics

xPos

```
private int xPos
```

Field Value

int

yPos

```
private int yPos
```

Field Value

int

Properties

PenColour

Gets or sets the current pen colour.

```
public object PenColour { get; set; }
```

Property Value

object

Xpos

Gets or sets the current X position of the drawing cursor.

```
public int Xpos { get; set; }
```

Property Value

int

Ypos

Gets or sets the current Y position of the drawing cursor.

```
public int Ypos { get; set; }
```

Property Value

int

Methods

Circle(int, bool)

Draws a circle from the current cursor position.

```
public void Circle(int radius, bool filled)
```

Parameters

`radius` int

Radius of the circle.

`filled` bool

Determines whether the circle is filled.

Clear()

Clears the canvas to white.

```
public void Clear()
```

DrawTo(int, int)

Draws a line from the current cursor position to the given coordinates.

```
public void DrawTo(int x, int y)
```

Parameters

`x` int

Target X coordinate.

`y` int

Target Y coordinate.

MoveTo(int, int)

Moves the cursor to a new position without drawing.

```
public void MoveTo(int x, int y)
```

Parameters

`x` int

New X coordinate.

`y` int

New Y coordinate.

Rect(int, int, bool)

Draws a rectangle from the current cursor position.

```
public void Rect(int width, int height, bool filled)
```

Parameters

`width` int

Width of the rectangle.

`height` int

Height of the rectangle.

`filled` bool

Determines whether the rectangle is filled.

Reset()

Resets the cursor to default position.

```
public void Reset()
```

Set(int, int)

Resizes the canvas.

```
public void Set(int width, int height)
```

Parameters

`width` int

New width.

`height` int

New height.

SetColour(int, int, int)

Sets the drawing colour using RGB values.

```
public void SetColour(int red, int green, int blue)
```

Parameters

`red` int

Red component.

`green` int

Green component.

`blue` int

Blue component.

Tri(int, int)

Draws a triangle using the current cursor position. The triangle can be drawn as an outline or filled shape.

```
public void Tri(int width, int height)
```

Parameters

`width` int

Width of the triangle.

`height` int

Height of the triangle.

WriteText(string)

Writes text to the canvas at the current cursor position. Supports plain text including symbols like £, $, %.

```
public void WriteText(string text)
```

Parameters

`text` string

Text to display.

getBitmap()

Returns the bitmap representing the canvas.

```
public object getBitmap()
```

Returns

object

Bitmap object.

Class AppCommandFactory

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Custom command factory used to create BOOSE commands. It extends the default command factory and returns application-specific command implementations.

```
public class AppCommandFactory : CommandFactory, ICommandFactory
```

Inheritance

object ← CommandFactory ← AppCommandFactory

Implements

ICommandFactory

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AppCommandFactory(ICanvas)

Initializes a new instance of the AppCommandFactory class.

```
public AppCommandFactory(ICanvas canvas)
```

Parameters

`canvas` ICanvas

Canvas used by commands created by the factory.

Fields

canvas

```
private ICanvas canvas
```

Field Value

ICanvas

Methods

MakeCommand(string)

Creates a command object based on the command name.

```
public override ICommand MakeCommand(string commandType)
```

Parameters

`commandType` string

The name of the BOOSE command.

Returns

ICommand

An BOOSE.ICommand implementation for the requested command.

Class ApplicationConfiguration

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Bootstrap the application configuration.

```
[CompilerGenerated]
internal static class ApplicationConfiguration
```

Inheritance

object ← ApplicationConfiguration

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Initialize()

Bootstrap the application as follows:

```
global::System.Windows.Forms.Application.EnableVisualStyles();
global::System.Windows.Forms.Application.SetCompatibleTextRenderingDefault(false);
global::System.Windows.Forms.Application.SetHighDpiMode(HighDpiMode.SystemAware);
```

```
public static void Initialize()
```

Class Array

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Array interpreter extension for BOOSE, adding support for array variables. Supports int variables, real variables, int arrays, real arrays, together with poke, peek and simple write commands.

```
public class Array : IInterpreter
```

Inheritance

object ← Array

Implements

IInterpreter

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Array(AppCanvas)

Initializes the array interpreter with a drawing canvas.

```
public Array(AppCanvas canvas)
```

Parameters

`canvas` AppCanvas

Canvas used for drawing operations.

Fields

addFeatures

```
private AdditionalFeatures addFeatures
```

Field Value

AdditionalFeatures

boolVars

```
private Dictionary<string, bool> boolVars
```

Field Value

Dictionary<string, bool>

canvas

```
private AppCanvas canvas
```

Field Value

AppCanvas

ifHandler

```
private If ifHandler
```

Field Value

If

intArrays

```
private Dictionary<string, int[]> intArrays
```

Field Value

Dictionary<string, int[]>

intVars

```
private Dictionary<string, int> intVars
```

Field Value

Dictionary<string, int>

maths

```
private Maths maths
```

Field Value

Maths

realArrays

```
private Dictionary<string, double[]> realArrays
```

Field Value

Dictionary<string, double[]>

realVars

```
private Dictionary<string, double> realVars
```

Field Value

Dictionary <string, double>

Methods

BuildWriteText(string)

Builds text for the write command. Supports: write x write y write "hello world!" write "" + x/y/z

```
private string BuildWriteText(string expr)
```

Parameters

`expr` string

Expression after write.

Returns

string

Text to display.

EvaluateBoolean(string)

Evaluates boolean expressions like: true, false

```
private bool EvaluateBoolean(string expr)
```

Parameters

`expr` string

Returns

bool

FindIntArrayName(string)

Methods that AI suggested to help out with unrestricet issue

```
private string? FindIntArrayName(string name)
```

Parameters

`name` string

Name of variable or array to find.

Returns

string

FindRealArrayName(string)

```
private string? FindRealArrayName(string name)
```

Parameters

`name` string

Returns

string

GetWriteValue(string)

Resolves value for write command. Can return string literal, int variable, real variable, boolean variable or plain text.

```
private string GetWriteValue(string value)
```

Parameters

`value` string

Input text or variable name.

Returns

string

Resolved text.

Run(string)

Executes a BOOSE program with array support.

```
public void Run(string program)
```

Parameters

`program` string

Program text entered by the user.

Class Circle

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

BOOSE command that draws a circle on the canvas.

```
internal class Circle : CommandOneParameter, ICommand
```

Inheritance

object ← Command ← CanvasCommand ← CommandOneParameter ← Circle

Implements

ICommand

Inherited Members

CommandOneParameter.param1 , CommandOneParameter.param1unprocessed , CommandOneParameter.CheckParameters(string[]) , CanvasCommand.yPos , CanvasCommand.xPos , CanvasCommand.canvas , CanvasCommand.Canvas , Command.IsDouble , Command.program , Command.parameterList , Command.parameters , Command.paramsint , Command.Set(StoredProgram, string) , Command.Compile() , Command.ProcessParameters(string) , Command.ToString() , Command.Program , Command.Name , Command.ParameterList , Command.Parameters , Command.Paramsint , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

Circle()

Default constructor used by the command factory.

```
public Circle()
```

Circle(Canvas, int)

Initializes a new instance of the Circle command.

```
public Circle(Canvas c, int radius)
```

Parameters

`c` Canvas

Canvas used for drawing.

`radius` int

Radius of the circle.

Fields

radius

```
private int radius
```

Field Value

int

Methods

Execute()

Executes the circle command and draws the circle.

```
public override void Execute()
```

Exceptions

CanvasException

Thrown when the radius parameter is invalid.

Class Clear

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Clears the drawing canvas.

```
internal class Clear : CommandOneParameter, ICommand
```

Inheritance

object ← Command ← CanvasCommand ← CommandOneParameter ← Clear

Implements

ICommand

Inherited Members

CommandOneParameter.param1 , CommandOneParameter.param1unprocessed , CommandOneParameter.CheckParameters(string[]) , CanvasCommand.yPos , CanvasCommand.xPos , CanvasCommand.canvas , CanvasCommand.Canvas , Command.IsDouble , Command.program , Command.parameterList , Command.parameters , Command.paramsint , Command.Set(StoredProgram, string) , Command.Compile() , Command.ProcessParameters(string) , Command.ToString() , Command.Program , Command.Name , Command.ParameterList , Command.Parameters , Command.Paramsint , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

Clear()

Initializes a new instance of the Clear command.

```
public Clear()
```

Methods

Execute()

Executes the clear command and clears the canvas.

```
public override void Execute()
```

Class Form1

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Main user interface for the BOOSE interpreter. Allows the user to input BOOSE programs, execute them, and display graphical output on the canvas.

```
public class Form1 : Form, IOleControl.Interface, IOleObject.Interface,
IOleInPlaceObject.Interface, IOleInPlaceActiveObject.Interface, IOleWindow.Interface,
IViewObject2.Interface, IViewObject.Interface, IPersistStreamInit.Interface,
IPersistPropertyBag.Interface, IPersistStorage.Interface, IPersist.Interface,
IQuickActivate.Interface, ISupportOleDropSource, IDropTarget, ISynchronizeInvoke,
IWin32Window, IBindableComponent, IKeyboardToolTip, IHandle<HWND>, IArrangedElement,
IComponent, IDisposable, IContainerControl
```

Inheritance

object ← MarshalByRefObject ← Component ← Control ← ScrollableControl ← ContainerControl ← Form ← Form1

Implements

IOleControl.Interface, IOleObject.Interface, IOleInPlaceObject.Interface, IOleInPlaceActiveObject.Interface, IOleWindow.Interface, IViewObject2.Interface, IViewObject.Interface, IPersistStreamInit.Interface, IPersistPropertyBag.Interface, IPersistStorage.Interface, IPersist.Interface, IQuickActivate.Interface, ISupportOleDropSource, IDropTarget, ISynchronizeInvoke, IWin32Window, IBindableComponent, IKeyboardToolTip, IHandle<HWND>, IArrangedElement, IComponent, IDisposable, IContainerControl

Inherited Members

Form.SetVisibleCore(bool) , Form.OnFormCornerPreferenceChanged(EventArgs) , Form.OnFormBorderColorChanged(EventArgs) , Form.OnFormCaptionBackColorChanged(EventArgs) , Form.OnFormCaptionTextColorChanged(EventArgs) , Form.Activate() , Form.ActivateMdiChild(Form) , Form.AddOwnedForm(Form) , Form.AdjustFormScrollbars(bool) , Form.ApplyAutoScaling() , Form.AssignParent(Control) , Form.Close() , Form.CreateAccessibilityInstance() , Form.CreateControlsInstance() , Form.AfterControlRemoved(Control, Control) , Form.CreateHandle() , Form.DefWndProc(ref Message) , Form.FocusInternal() , Form.GetAutoScaleSize(Font) , Form.CanSelectCore() , Form.CanProcessMnemonic() , Form.ProcessMnemonic(char) , Form.CenterToParent() , Form.CenterToScreen() , Form.LayoutMdi(MdiLayout) ,

Form.OnActivated(EventArgs) , Form.OnAutoScaleModeChanged() ,
Form.OnBackgroundImageChanged(EventArgs) ,
Form.OnBackgroundImageLayoutChanged(EventArgs) , Form.OnClosing(CancelEventArgs) ,
Form.OnClosed(EventArgs) , Form.OnFormClosing(FormClosingEventArgs) ,
Form.OnFormClosed(FormClosedEventArgs) , Form.OnCreateControl() ,
Form.OnDeactivate(EventArgs) , Form.OnEnabledChanged(EventArgs) , Form.OnEnter(EventArgs) ,
Form.OnFontChanged(EventArgs) , Form.OnGotFocus(EventArgs) ,
Form.OnHandleCreated(EventArgs) , Form.OnHandleDestroyed(EventArgs) ,
Form.OnHelpButtonClicked(CancelEventArgs) , Form.OnLayout(LayoutEventArgs) ,
Form.OnLoad(EventArgs) , Form.OnMaximizedBoundsChanged(EventArgs) ,
Form.OnMaximumSizeChanged(EventArgs) , Form.OnMinimumSizeChanged(EventArgs) ,
Form.OnInputLanguageChanged(InputLanguageChangedEventArgs) ,
Form.OnInputLanguageChanging(InputLanguageChangingEventArgs) ,
Form.OnVisibleChanged(EventArgs) , Form.OnMdiChildActivate(EventArgs) ,
Form.OnMenuStart(EventArgs) , Form.OnMenuComplete(EventArgs) ,
Form.OnPaint(PaintEventArgs) , Form.OnResize(EventArgs) ,
Form.OnDpiChanged(DpiChangedEventArgs) , Form.OnGetDpiScaledSize(int, int, ref Size) ,
Form.OnRightToLeftLayoutChanged(EventArgs) , Form.OnShown(EventArgs) ,
Form.OnTextChanged(EventArgs) , Form.ProcessCmdKey(ref Message, Keys) ,
Form.ProcessDialogKey(Keys) , Form.ProcessDialogChar(char) ,
Form.ProcessKeyPreview(ref Message) , Form.ProcessTabKey(bool) , Form.RecreateHandleCore() ,
Form.RemoveOwnedForm(Form) , Form.Select(bool, bool) , Form.ScaleCore(float, float) ,
Form.ScaleMinMaxSize(float, float, bool) ,
Form.GetScaledBounds(Rectangle, SizeF, BoundsSpecified) ,
Form.ScaleControl(SizeF, BoundsSpecified) , Form.SetBoundsCore(int, int, int, int, BoundsSpecified) ,
Form.SetClientSizeCore(int, int) , Form.SetDesktopBounds(int, int, int, int) ,
Form.SetDesktopLocation(int, int) , Form.Show(IWin32Window) ,
Form.ShowAsync(IWin32Window) , Form.SuppressApplicationOnThreadException(Exception) ,
Form.ShowDialog() , Form.ShowDialog(IWin32Window) , Form.ShowDialogAsync() ,
Form.ShowDialogAsync(IWin32Window) , Form.ShouldSerializeAutoScaleBaseSize() ,
Form.ShouldSerializeSize() , Form.ToString() , Form.UpdateDefaultButton() ,
Form.OnResizeBegin(EventArgs) , Form.OnResizeEnd(EventArgs) ,
Form.OnStyleChanged(EventArgs) , Form.ValidateChildren() ,
Form.ValidateChildren(ValidationConstraints) , Form.WndProc(ref Message) , Form.AcceptButton ,
Form.ActiveForm , Form.ActiveMdiChild , Form.AllowTransparency , Form.AutoScale ,
Form.AutoScaleBaseSize , Form.AutoScroll , Form.AutoSize , Form.AutoSizeMode ,
Form.AutoValidate , Form.BackColor , Form.FormBorderStyle , Form.CancelButton ,
Form.ClientSize , Form.ControlBox , Form.CreateParams , Form.DefaultImeMode ,
Form.DefaultSize , Form.DesktopBounds , Form.DesktopLocation , Form.DialogResult ,
Form.HelpButton , Form.Icon , Form.IsMdiChild , Form.IsMdiContainer ,

Form.IsRestrictedWindow , Form.KeyPreview , Form.Location , Form.MaximizedBounds , Form.MaximumSize , Form.MainMenuStrip , Form.Margin , Form.MinimumSize , Form.MaximizeBox , Form.MdiChildren , Form.MdiChildrenMinimizedAnchorBottom , Form.MdiParent , Form.MinimizeBox , Form.Modal , Form.Opacity , Form.OwnedForms , Form.Owner , Form.RestoreBounds , Form.RightToLeftLayout , Form.ParentInternal , Form.ShowInTaskbar , Form.ShowIcon , Form.ShowParams , Form.ShowWithoutActivation , Form.Size , Form.SizeGripStyle , Form.StartPosition , Form.TabIndex , Form.TabStop , Form.Text , Form.TopLevel , Form.TopMost , Form.TransparencyKey , Form.FormCornerPreference , Form.FormBorderColor , Form.FormCaptionBackColor , Form.FormCaptionTextColor , Form.WindowState , Form.WindowText , Form.SupportsUiaProviders , Form.AutoSizeChanged , Form.AutoValidateChanged , Form.HelpButtonClicked , Form.MaximizedBoundsChanged , Form.MaximumSizeChanged , Form.MarginChanged , Form.MinimumSizeChanged , Form.TabIndexChanged , Form.TabStopChanged , Form.Activated , Form.Closing , Form.Closed , Form.Deactivate , Form.FormClosing , Form.FormBorderColorChanged , Form.FormCaptionBackColorChanged , Form.FormCaptionTextColorChanged , Form.FormCornerPreferenceChanged , Form.FormClosed , Form.Load , Form.MdiChildActivate , Form.MenuComplete , Form.MenuStart , Form.InputLanguageChanged , Form.InputLanguageChanging , Form.RightToLeftLayoutChanged , Form.Shown , Form.DpiChanged , Form.ResizeBegin , Form.ResizeEnd , ContainerControl.IContainerControl.ActivateControl(Control) , ContainerControl.GetPreferredSizeCore(Size) , ContainerControl.GetToolNativeScreenRectangle() , ContainerControl.OnAutoValidateChanged(EventArgs) , ContainerControl.OnFrameWindowActivate(bool) , ContainerControl.OnChildLayoutResuming(Control, bool) , ContainerControl.OnLayoutResuming(bool) , ContainerControl.OnMove(EventArgs) , ContainerControl.OnParentChanged(EventArgs) , ContainerControl.PerformAutoScale() , ContainerControl.Scale(SizeF, SizeF, Control, bool) , ContainerControl.RescaleConstantsForDpi(int, int) , ContainerControl.Validate() , ContainerControl.Validate(bool) , ContainerControl.ShouldSerializeAutoValidate() , ContainerControl.AutoScaleDimensions , ContainerControl.AutoScaleFactor , ContainerControl.AutoScaleMode , ContainerControl.BindingContext , ContainerControl.CanEnableIme , ContainerControl.ActiveControl , ContainerControl.CurrentAutoScaleDimensions , ContainerControl.ParentForm , ContainerControl.InnerMostActiveContainerControl , ScrollableControl.ScrollStateAutoScrolling , ScrollableControl.ScrollStateHScrollVisible , ScrollableControl.ScrollStateVScrollVisible , ScrollableControl.ScrollStateUserHasScrolled , ScrollableControl.ScrollStateFullDrag , ScrollableControl.GetScrollState(int) , ScrollableControl.OnMouseWheel(MouseEventArgs) , ScrollableControl.OnRightToLeftChanged(EventArgs) , ScrollableControl.OnPaintBackground(PaintEventArgs) , ScrollableControl.OnPaddingChanged(EventArgs) , ScrollableControl.SetDisplayRectLocation(int, int) ,

ScrollableControl.ScrollControlIntoView(Control) , ScrollableControl.ScrollToControl(Control) ,
ScrollableControl.OnScroll(ScrollEventArgs) , ScrollableControl.SetAutoScrollMargin(int, int) ,
ScrollableControl.SetScrollState(int, bool) , ScrollableControl.AutoScrollMargin ,
ScrollableControl.AutoScrollPosition , ScrollableControl.AutoScrollMinSize ,
ScrollableControl.DisplayRectangle , ScrollableControl.IArrangedElement.DisplayRectangle ,
ScrollableControl.HScroll , ScrollableControl.HorizontalScroll , ScrollableControl.VScroll ,
ScrollableControl.VerticalScroll , ScrollableControl.DockPadding , ScrollableControl.Scroll ,
Control.s_paddingChangedEvent , Control.IOleControl.Interface.GetControlInfo(CONTROLINFO*) ,
Control.IOleControl.Interface.OnMnemonic(MSG*) ,
Control.IOleControl.Interface.OnAmbientPropertyChange(int) ,
Control.IOleControl.Interface.FreezeEvents(BOOL) ,
Control.IOleInPlaceActiveObject.Interface.GetWindow(HWND*) ,
Control.IOleInPlaceActiveObject.Interface.ContextSensitiveHelp(BOOL) ,
Control.IOleInPlaceActiveObject.Interface.TranslateAccelerator(MSG*) ,
Control.IOleInPlaceActiveObject.Interface.OnFrameWindowActivate(BOOL) ,
Control.IOleInPlaceActiveObject.Interface.OnDocWindowActivate(BOOL) ,
Control.IOleInPlaceActiveObject.Interface.ResizeBorder(RECT*, IOleInPlaceUIWindow*, BOOL) ,
Control.IOleInPlaceActiveObject.Interface.EnableModeless(BOOL) ,
Control.IOleInPlaceObject.Interface.GetWindow(HWND*) ,
Control.IOleInPlaceObject.Interface.ContextSensitiveHelp(BOOL) ,
Control.IOleInPlaceObject.Interface.InPlaceDeactivate() ,
Control.IOleInPlaceObject.Interface.UIDeactivate() ,
Control.IOleInPlaceObject.Interface.SetObjectRects(RECT*, RECT*) ,
Control.IOleInPlaceObject.Interface.ReactivateAndUndo() ,
Control.IOleObject.Interface.SetClientSite(IOleClientSite*) ,
Control.IOleObject.Interface.GetClientSite(IOleClientSite**) ,
Control.IOleObject.Interface.SetHostNames(PCWSTR, PCWSTR) ,
Control.IOleObject.Interface.Close(uint) , Control.IOleObject.Interface.SetMoniker(uint, IMoniker*) ,
Control.IOleObject.Interface.GetMoniker(uint, uint, IMoniker**) ,
Control.IOleObject.Interface.InitFromData(IDataObject*, BOOL, uint) ,
Control.IOleObject.Interface.GetClipboardData(uint, IDataObject**) ,
Control.IOleObject.Interface.DoVerb(int, MSG*, IOleClientSite*, int, HWND, RECT*) ,
Control.IOleObject.Interface.EnumVerbs(IEnumOLEVERB**) , Control.IOleObject.Interface.Update() ,
Control.IOleObject.Interface.IsUpToDate() , Control.IOleObject.Interface.GetUserClassID(Guid*) ,
Control.IOleObject.Interface.GetUserType(uint, PWSTR*) ,
Control.IOleObject.Interface.SetExtent(DVASPECT, SIZE*) ,
Control.IOleObject.Interface.GetExtent(DVASPECT, SIZE*) ,
Control.IOleObject.Interface.Advise(IAdviseSink*, uint*) , Control.IOleObject.Interface.Unadvise(uint) ,
Control.IOleObject.Interface.EnumAdvise(IEnumSTATDATA**) ,
Control.IOleObject.Interface.GetMiscStatus(DVASPECT, OLEMISC*) ,

Control.IOleObject.Interface.SetColorScheme(LOGPALETTE*) ,
Control.IOleWindow.Interface.GetWindow(HWND*) ,
Control.IOleWindow.Interface.ContextSensitiveHelp(BOOL) ,
Control.IPersist.Interface.GetClassID(Guid*) , Control.IPersistPropertyBag.Interface.InitNew() ,
Control.IPersistPropertyBag.Interface.GetClassID(Guid*) ,
Control.IPersistPropertyBag.Interface.Load(IPropertyBag*, IErrorLog*) ,
Control.IPersistPropertyBag.Interface.Save(IPropertyBag*, BOOL, BOOL) ,
Control.IPersistStorage.Interface.GetClassID(Guid*) , Control.IPersistStorage.Interface.IsDirty() ,
Control.IPersistStorage.Interface.InitNew(IStorage*) ,
Control.IPersistStorage.Interface.Load(IStorage*) ,
Control.IPersistStorage.Interface.Save(IStorage*, BOOL) ,
Control.IPersistStorage.Interface.SaveCompleted(IStorage*) ,
Control.IPersistStorage.Interface.HandsOffStorage() ,
Control.IPersistStreamInit.Interface.GetClassID(Guid*) , Control.IPersistStreamInit.Interface.IsDirty() ,
Control.IPersistStreamInit.Interface.Load(IStream*) ,
Control.IPersistStreamInit.Interface.Save(IStream*, BOOL) ,
Control.IPersistStreamInit.Interface.GetSizeMax(ulong*) ,
Control.IPersistStreamInit.Interface.InitNew() ,
Control.IQuickActivate.Interface.QuickActivate(QACONTAINER*, QACONTROL*) ,
Control.IQuickActivate.Interface.SetContentExtent(SIZE*) ,
Control.IQuickActivate.Interface.GetContentExtent(SIZE*) ,
Control.IViewObject.Interface.Draw(DVASPECT, int, void*, DVTARGETDEVICE*, HDC, HDC, RECTL*, RECTL*,
nint, nuint) ,
Control.IViewObject.Interface.GetColorSet(DVASPECT, int, void*, DVTARGETDEVICE*, HDC,
LOGPALETTE**) ,
Control.IViewObject.Interface.Freeze(DVASPECT, int, void*, uint*) ,
Control.IViewObject.Interface.Unfreeze(uint) ,
Control.IViewObject.Interface.SetAdvise(DVASPECT, uint, IAdviseSink*) ,
Control.IViewObject.Interface.GetAdvise(uint*, uint*, IAdviseSink**) ,
Control.IViewObject2.Interface.Draw(DVASPECT, int, void*, DVTARGETDEVICE*, HDC, HDC, RECTL*,
RECTL*, nint, nuint) ,
Control.IViewObject2.Interface.GetColorSet(DVASPECT, int, void*, DVTARGETDEVICE*, HDC,
LOGPALETTE**) ,
Control.IViewObject2.Interface.Freeze(DVASPECT, int, void*, uint*) ,
Control.IViewObject2.Interface.Unfreeze(uint) ,
Control.IViewObject2.Interface.SetAdvise(DVASPECT, uint, IAdviseSink*) ,
Control.IViewObject2.Interface.GetAdvise(uint*, uint*, IAdviseSink**) ,
Control.IViewObject2.Interface.GetExtent(DVASPECT, int, DVTARGETDEVICE*, SIZE*) ,
Control.GetAccessibilityObjectById(int) , Control.SetAutoSizeMode(AutoSizeMode) ,
Control.GetAutoSizeMode() , Control.ResetBindings() , Control.GetPreferredSize(Size) ,

Control.IsHoveredWithMouse() , Control.AddReflectChild() , Control.RemoveReflectChild() ,
Control.RemoveToolTip(ToolTip) , Control.AccessibilityNotifyClients(AccessibleEvents, int) ,
Control.AccessibilityNotifyClients(AccessibleEvents, int, int) , Control.BeginInvoke(Delegate) ,
Control.BeginInvoke(Action) , Control.BeginInvoke(Delegate, params object[]) ,
Control.BringToFront() , Control.Contains(Control) , Control.CreateGraphics() ,
Control.CreateControl() , Control.DestroyHandle() , Control.DisposeAxControls() ,
Control.DoDragDrop(object, DragDropEffects) ,
Control.DoDragDrop(object, DragDropEffects, Bitmap, Point, bool) ,
Control.DrawToBitmap(Bitmap, Rectangle) , Control.EndInvoke(IAsyncResult) , Control.FindForm() ,
Control.GetTopLevel() , Control.RaiseKeyEvent(object, KeyEventArgs) ,
Control.RaiseMouseEvent(object, MouseEventArgs) , Control.Focus() ,
Control.FromChildHandle(nint) , Control.FromHandle(nint) ,
Control.ApplyBoundsConstraints(int, int, int, int) ,
Control.GetChildAtPoint(Point, GetChildAtPointSkip) , Control.GetCaptionForTool(ToolTip) ,
Control.GetChildAtPoint(Point) , Control.GetContainerControl() ,
Control.GetFirstChildControlInTabOrder(bool) , Control.GetNeighboringToolsRectangles() ,
Control.GetNextControl(Control, bool) , Control.GetState(Control.States) ,
Control.GetExtendedState(Control.ExtendedStates) , Control.GetStyle(ControlStyles) ,
Control.Hide() , Control.InitLayout() , Control.InitializeDCForWmCtlColor(HDC, MessageId) ,
Control.Invalidate(Region) , Control.Invalidate(Region, bool) , Control.Invalidate() ,
Control.Invalidate(bool) , Control.Invalidate(Rectangle) , Control.Invalidate(Rectangle, bool) ,
Control.Invoke(Action) , Control.Invoke(Delegate) , Control.Invoke(Delegate, params object[]) ,
Control.Invoke<T>(Func<T>) , Control.InvokePaint(Control, PaintEventArgs) ,
Control.InvokePaintBackground(Control, PaintEventArgs) , Control.IsKeyLocked(Keys) ,
Control.IsInputChar(char) , Control.IsInputKey(Keys) , Control.IsMnemonic(char, string) ,
Control.LogicalToDeviceUnits(int) , Control.LogicalToDeviceUnits(Size) ,
Control.ScaleBitmapLogicalToDevice(ref Bitmap) ,
Control.AdjustWindowRectExForControlDpi(ref RECT, WINDOW_STYLE, bool, WINDOW_EX_STYLE) ,
Control.NotifyInvalidate(Rectangle) , Control.InvokeOnClick(Control, EventArgs) ,
Control.OnAutoSizeChanged(EventArgs) , Control.OnBackColorChanged(EventArgs) ,
Control.OnBindingContextChanged(EventArgs) , Control.OnCausesValidationChanged(EventArgs) ,
Control.OnContextMenuStripChanged(EventArgs) , Control.OnCursorChanged(EventArgs) ,
Control.OnDataContextChanged(EventArgs) , Control.OnDockChanged(EventArgs) ,
Control.OnForeColorChanged(EventArgs) , Control.OnNotifyMessage(Message) ,
Control.OnParentBackColorChanged(EventArgs) ,
Control.OnParentBackgroundImageChanged(EventArgs) ,
Control.OnParentBindingContextChanged(EventArgs) , Control.OnParentCursorChanged(EventArgs) ,
Control.OnParentDataContextChanged(EventArgs) , Control.OnParentEnabledChanged(EventArgs) ,
Control.OnParentFontChanged(EventArgs) , Control.OnParentHandleRecreated() ,
Control.OnParentHandleRecreating() , Control.OnParentForeColorChanged(EventArgs) ,

Control.OnParentRightToLeftChanged(EventArgs) , Control.OnParentVisibleChanged(EventArgs) ,
Control.OnParentBecameInvisible() , Control.OnPrint(PaintEventArgs) ,
Control.OnTabIndexChanged(EventArgs) , Control.OnTabStopChanged(EventArgs) ,
Control.OnTopMostActiveXParentChanged(EventArgs) , Control.OnClick(EventArgs) ,
Control.OnClientSizeChanged(EventArgs) , Control.OnControlAdded(ControlEventArgs) ,
Control.OnControlRemoved(ControlEventArgs) ,
Control.OnInvokedSetScrollPosition(object, EventArgs) , Control.OnLocationChanged(EventArgs) ,
Control.OnDoubleClick(EventArgs) , Control.OnDragEnter(DragEventArgs) ,
Control.OnDragOver(DragEventArgs) , Control.OnDragLeave(EventArgs) ,
Control.OnDragDrop(DragEventArgs) , Control.OnGiveFeedback(GiveFeedbackEventArgs) ,
Control.InvokeGotFocus(Control, EventArgs) , Control.OnHelpRequested(HelpEventArgs) ,
Control.OnInvalidated(InvalidateEventArgs) , Control.OnKeyDown(KeyEventArgs) ,
Control.OnKeyPress(KeyPressEventArgs) , Control.OnKeyUp(KeyEventArgs) ,
Control.OnLayoutSuspended() , Control.OnLeave(EventArgs) ,
Control.InvokeLostFocus(Control, EventArgs) , Control.OnLostFocus(EventArgs) ,
Control.OnMarginChanged(EventArgs) , Control.OnMouseDoubleClick(MouseEventArgs) ,
Control.OnMouseClick(MouseEventArgs) , Control.OnMouseCaptureChanged(EventArgs) ,
Control.OnMouseDown(MouseEventArgs) , Control.OnMouseEnter(EventArgs) ,
Control.OnMouseLeave(EventArgs) , Control.OnDpiChangedBeforeParent(EventArgs) ,
Control.OnDpiChangedAfterParent(EventArgs) , Control.OnMouseHover(EventArgs) ,
Control.OnMouseMove(MouseEventArgs) , Control.OnMouseUp(MouseEventArgs) ,
Control.OnQueryContinueDrag(QueryContinueDragEventArgs) ,
Control.OnRegionChanged(EventArgs) , Control.OnPreviewKeyDown(PreviewKeyDownEventArgs) ,
Control.OnSizeChanged(EventArgs) , Control.OnChangeUICues(UICuesEventArgs) ,
Control.OnSystemColorsChanged(EventArgs) , Control.OnValidating(CancelEventArgs) ,
Control.OnValidated(EventArgs) , Control.InitializeConstantsForInitialDpi(int) ,
Control.PerformLayout() , Control.PerformLayout(Control, string) , Control.PointToClient(Point) ,
Control.PointToScreen(Point) , Control.PreProcessMessage(ref Message) ,
Control.PreProcessControlMessage(ref Message) ,
Control.PrintToMetaFileRecursive(HDC, nint, Rectangle) , Control.ProcessKeyEventArgs(ref Message) ,
Control.ProcessKeyMessage(ref Message) , Control.RaiseDragEvent(object, DragEventArgs) ,
Control.RaisePaintEvent(object, PaintEventArgs) , Control.ResetBackColor() , Control.ResetCursor() ,
Control.ResetFont() , Control.ResetForeColor() , Control.ResetRightToLeft() ,
Control.RecreateHandle() , Control.RectangleToClient(Rectangle) ,
Control.RectangleToScreen(Rectangle) , Control.ReflectMessage(nint, ref Message) ,
Control.Refresh() , Control.ReleaseUiaProvider(HWND) ,
Control.TryGetAccessibilityObject(out AccessibleObject) , Control.ResetMouseEventArgs() ,
Control.ResetText() , Control.ResumeLayout() , Control.ResumeLayout(bool) , Control.Scale(float) ,
Control.Scale(float, float) , Control.Scale(SizeF) , Control.Select() ,
Control.SelectNextControl(Control, bool, bool, bool, bool) , Control.SendToBack() ,

Control.SetBounds(int, int, int, int) , Control.SetBounds(int, int, int, int, BoundsSpecified) ,
Control.SizeFromClientSize(Size) , Control.SetState(Control.States, bool) ,
Control.SetExtendedState(Control.ExtendedStates, bool) , Control.SetStyle(ControlStyles, bool) ,
Control.SetToolTip(ToolTip) , Control.SetTopLevel(bool) , Control.SetTopLevelInternal(bool) ,
Control.ShouldPerformContainerValidation() , Control.ShouldSerializeBackColor() ,
Control.ShouldSerializeCursor() , Control.ShouldSerializeForeColor() , Control.ShouldSerializeFont() ,
Control.ShouldSerializeRightToLeft() , Control.RtlTranslateAlignment(HorizontalAlignment) ,
Control.RtlTranslateAlignment(LeftRightAlignment) ,
Control.RtlTranslateAlignment(ContentAlignment) ,
Control.RtlTranslateHorizontal(HorizontalAlignment) ,
Control.RtlTranslateLeftRight(LeftRightAlignment) , Control.RtlTranslateContent(ContentAlignment) ,
Control.Show() , Control.ShouldSerializeMaximumSize() , Control.ShouldSerializeMinimumSize() ,
Control.ShouldSerializeText() , Control.SuspendLayout() , Control.TryGetExplicitlySetFont(out Font) ,
Control.SetScaledFont(Font, bool) , Control.Update() , Control.UpdateBounds() ,
Control.UpdateBounds(int, int, int, int) , Control.UpdateBounds(int, int, int, int, int, int) ,
Control.UpdateZOrder() , Control.UpdateStyles() , Control.UpdateStylesCore() ,
Control.OnBoundsUpdate(int, int, int, int) , Control.WmContextMenu(ref Message) ,
Control.IArrangedElement.PerformLayout(IArrangedElement, string) ,
Control.IArrangedElement.SetBounds(Rectangle, BoundsSpecified) ,
Control.IDropTarget.OnDragEnter(DragEventArgs) ,
Control.IDropTarget.OnDragOver(DragEventArgs) , Control.IDropTarget.OnDragLeave(EventArgs) ,
Control.IDropTarget.OnDragDrop(DragEventArgs) ,
Control.ISupportOleDropSource.OnGiveFeedback(GiveFeedbackEventArgs) ,
Control.ISupportOleDropSource.OnQueryContinueDrag(QueryContinueDragEventArgs) ,
Control.IKeyboardToolTip.CanShowToolTipsNow() ,
Control.IKeyboardToolTip.GetNativeScreenRectangle() ,
Control.IKeyboardToolTip.GetNeighboringToolsRectangles() ,
Control.IKeyboardToolTip.IsHoveredWithMouse() , Control.IKeyboardToolTip.HasRtlModeEnabled() ,
Control.IKeyboardToolTip.AllowsToolTip() , Control.IKeyboardToolTip.GetOwnerWindow() ,
Control.IKeyboardToolTip.OnHooked(ToolTip) , Control.IKeyboardToolTip.OnUnhooked(ToolTip) ,
Control.IKeyboardToolTip.GetCaptionForTool(ToolTip) ,
Control.IKeyboardToolTip.ShowsOwnToolTip() , Control.IKeyboardToolTip.IsBeingTabbedTo() ,
Control.IKeyboardToolTip.AllowsChildrenToShowToolTips() , Control.ShowsOwnKeyboardToolTip() ,
Control.OnKeyboardToolTipHook(ToolTip) , Control.OnKeyboardToolTipUnhook(ToolTip) ,
Control.AllowsKeyboardToolTip() , Control.GetToolInfoWrapper(TOOLTIP_FLAGS, string, ToolTip) ,
Control.AllowsChildrenToShowToolTips() , Control.OnImeModeChanged(EventArgs) ,
Control.ResetImeMode() , Control.ShouldSerializeImeMode() ,
Control.InvokeAsync(Action, CancellationToken) ,
Control.InvokeAsync<T>(Func<T>, CancellationToken) ,
Control.InvokeAsync(Func<CancellationToken, ValueTask>, CancellationToken) ,

Control.InvokeAsync<T>(Func<CancellationToken, ValueTask<T>>, CancellationToken) ,
Control.AccessibilityObject , Control.AccessibleDefaultActionDescription ,
Control.AccessibleDescription , Control.AccessibleName , Control.AccessibleRole ,
Control.AllowDrop , Control.Anchor , Control.AutoScrollOffset , Control.LayoutEngine ,
Control.DataContext , Control.BackgroundImage , Control.BackgroundImageLayout ,
Control.Bottom , Control.Bounds , Control.CanAccessProperties , Control.CanFocus ,
Control.CanRaiseEvents , Control.CanSelect , Control.Capture , Control.CausesValidation ,
Control.CheckForIllegalCrossThreadCalls , Control.ClientRectangle , Control.CompanyName ,
Control.ContainsFocus , Control.ContextMenuStrip , Control.Controls , Control.Created ,
Control.Cursor , Control.DataBindings , Control.DefaultBackColor , Control.DefaultCursor ,
Control.DefaultFont , Control.DefaultForeColor , Control.DefaultMargin ,
Control.DefaultMaximumSize , Control.DefaultMinimumSize , Control.DefaultPadding ,
Control.DeviceDpi , Control.IsDisposed , Control.Disposing , Control.Dock ,
Control.DoubleBuffered , Control.Enabled , Control.Focused , Control.Font ,
Control.FontHeight , Control.ForeColor , Control.Handle , Control.HasChildren , Control.Height ,
Control.IsHandleCreated , Control.InvokeRequired , Control.IsAccessible ,
Control.IsAncestorSiteInDesignMode , Control.IsContainerControl , Control.IsMirrored ,
Control.IsMnemonicsListenerAxSourced , Control.LastKeyData , Control.Left ,
Control.ModifierKeys , Control.MouseButtons , Control.MousePosition , Control.Name ,
Control.Parent , Control.ProductName , Control.ProductVersion , Control.RecreatingHandle ,
Control.Region , Control.RenderRightToLeft , Control.RenderTransparencyWithVisualStyles ,
Control.ResizeRedraw , Control.Right , Control.RightToLeft , Control.ScaleChildren , Control.Site ,
Control.Tag , Control.Top , Control.TopLevelControl , Control.ShowKeyboardCues ,
Control.ShowFocusCues , Control.UseWaitCursor , Control.SupportsUseCompatibleTextRendering ,
Control.Visible , Control.Width , Control.ExtendedWindowStyle , Control.WindowTarget ,
Control.PreferredSize , Control.Padding , Control.IArrangedElement.Children ,
Control.IArrangedElement.Container , Control.IArrangedElement.ParticipatesInLayout ,
Control.IArrangedElement.Properties , Control.IHandle<HWND>.Handle , Control.ImeMode ,
Control.ImeModeBase , Control.PropagatingImeMode , Control.BackColorChanged ,
Control.BackgroundImageChanged , Control.BackgroundImageLayoutChanged ,
Control.BindingContextChanged , Control.CausesValidationChanged , Control.ClientSizeChanged ,
Control.ContextMenuStripChanged , Control.CursorChanged , Control.DockChanged ,
Control.EnabledChanged , Control.FontChanged , Control.ForeColorChanged ,
Control.LocationChanged , Control.RegionChanged , Control.RightToLeftChanged ,
Control.SizeChanged , Control.TextChanged , Control.VisibleChanged , Control.Click ,
Control.ControlAdded , Control.ControlRemoved , Control.DataContextChanged ,
Control.DragDrop , Control.DragEnter , Control.DragOver , Control.DragLeave ,
Control.GiveFeedback , Control.HandleCreated , Control.HandleDestroyed ,
Control.HelpRequested , Control.Invalidated , Control.PaddingChanged , Control.Paint ,
Control.QueryContinueDrag , Control.QueryAccessibilityHelp , Control.DoubleClick ,

Control.Enter , Control.GotFocus , Control.KeyDown , Control.KeyPress , Control.KeyUp , Control.Layout , Control.Leave , Control.LostFocus , Control.MouseClick , Control.MouseDoubleClick , Control.MouseCaptureChanged , Control.MouseDown , Control.MouseEnter , Control.MouseLeave , Control.DpiChangedBeforeParent , Control.DpiChangedAfterParent , Control.MouseHover , Control.MouseMove , Control.MouseUp , Control.MouseWheel , Control.Move , Control.PreviewKeyDown , Control.Resize , Control.ChangeUICues , Control.StyleChanged , Control.SystemColorsChanged , Control.Validating , Control.Validated , Control.ParentChanged , Control.ImeModeChanged , Component.Dispose() , Component.GetService(Type) , Component.Container , Component.DesignMode , Component.Events , Component.Disposed , MarshalByRefObject.GetLifetimeService() , MarshalByRefObject.InitializeLifetimeService() , MarshalByRefObject.MemberwiseClone(bool) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

Form1()

Initializes the main form and sets up BOOSE components.

```
public Form1()
```

Fields

Factory

```
private CommandFactory Factory
```

Field Value

CommandFactory

Parser

```
private IParser Parser
```

Field Value

IParser

Program

```
private StoredProgram Program
```

Field Value

StoredProgram

button1

```
private Button button1
```

Field Value

Button

button2

```
private Button button2
```

Field Value

Button

canvas

```
private AppCanvas canvas
```

Field Value

AppCanvas

components

Required designer variable.

```
private IContainer components
```

Field Value

IContainer

pictureBox1

```
private PictureBox pictureBox1
```

Field Value

PictureBox

richTextBox1

```
private RichTextBox richTextBox1
```

Field Value

RichTextBox

textBox1

```
private TextBox textBox1
```

Field Value

TextBox

Properties

UserInput

Gets or sets the BOOSE program entered by the user.

```
[Browsable(false)]
[DesignerSerializationVisibility(DesignerSerializationVisibility.Hidden)]
public string UserInput { get; set; }
```

Property Value

string

Methods

Dispose(bool)

Clean up any resources being used.

```
protected override void Dispose(bool disposing)
```

Parameters

`disposing` bool

true if managed resources should be disposed; otherwise, false.

Form1_Load(object, EventArgs)

```
private void Form1_Load(object sender, EventArgs e)
```

Parameters

`sender` object

`e` EventArgs

GetInterpreter(string)

Selects the appropriate interpreter strategy based on the program content.

```
private IInterpreter GetInterpreter(string input)
```

Parameters

`input` string

Program entered by the user.

Returns

IInterpreter

An interpreter implementing the selected strategy.

InitializeComponent()

Required method for Designer support - do not modify the contents of this method with the code editor.

```
private void InitializeComponent()
```

button1_Click(object, EventArgs)

```
public void button1_Click(object sender, EventArgs e)
```

Parameters

`sender` object

`e` EventArgs

button2_Click(object, EventArgs)

```
private void button2_Click(object sender, EventArgs e)
```

Parameters

`sender` object

`e` EventArgs

panel1_Paint(object, PaintEventArgs)

```
private void panel1_Paint(object sender, PaintEventArgs e)
```

Parameters

`sender` object

`e` PaintEventArgs

pictureBox1_Click(object, EventArgs)

```
private void pictureBox1_Click(object sender, EventArgs e)
```

Parameters

`sender` object

`e` EventArgs

pictureBox1_Paint(object, PaintEventArgs)

```
private void pictureBox1_Paint(object sender, PaintEventArgs e)
```

Parameters

sender object

e PaintEventArgs

textBox1_TextChanged(object, EventArgs)

```
private void textBox1_TextChanged(object sender, EventArgs e)
```

Parameters

sender object

e EventArgs

textBox2_TextChanged(object, EventArgs)

```
private void textBox2_TextChanged(object sender, EventArgs e)
```

Parameters

sender object

e EventArgs

Interface IInterpreter

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Common interface for classes that can run a BOOSE program.

```
public interface IInterpreter
```

Methods

Run(string)

Runs the supplied BOOSE program.

```
void Run(string program)
```

Parameters

`program` string

Program entered by the user.

Class If

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Handles IF, ELSE and END IF logic in BOOSE programs.

```
public class If
```

Inheritance

object ← If

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

If(Maths)

```
public If(Maths maths)
```

Parameters

`maths` Maths

Fields

maths

```
private Maths maths
```

Field Value

Maths

Methods

Evaluate(string)

Evaluates conditions like: `x < 10`, `x > 5`, `x == 3`.

```
public bool Evaluate(string cond)
```

Parameters

`cond` string

Returns

bool

Class Int

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Simple interpreter extension that adds support for integer variables (int). Allows declaration, assignment and usage of variables in BOOSE commands.

```
public class Int : IInterpreter
```

Inheritance

object ← Int

Implements

IInterpreter

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Int(AppCanvas)

Initializes the interpreter with a drawing canvas.

```
public Int(AppCanvas canvas)
```

Parameters

`canvas` AppCanvas

Canvas used for drawing operations.

Fields

canvas

```
private AppCanvas canvas
```

Field Value

AppCanvas

maths

```
private Maths maths
```

Field Value

Maths

vars

```
private Dictionary<string, int> vars
```

Field Value

Dictionary <string, int>

Methods

Run(string)

Executes a BOOSE program with integer variable support.

```
public void Run(string program)
```

Parameters

`program` string

Program text entered by the user.

Class LogInDebug

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Singleton logger used to write debug information.

```
public sealed class LogInDebug
```

Inheritance

object ← LogInDebug

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

LogInDebug()

```
private LogInDebug()
```

Fields

instance

```
private static LogInDebug instance
```

Field Value

LogInDebug

Properties

Instance

Gets the single logger instance.

```
public static LogInDebug Instance { get; }
```

Property Value

LogInDebug

Methods

Log(string)

Writes a message to the debug output.

```
public void Log(string message)
```

Parameters

`message` string

Class Maths

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Provides simple mathematical expression evaluation (+, -, *, /)

```
public class Maths
```

Inheritance

object ← Maths

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Maths(Dictionary<string, int>, Dictionary<string, double>)

Initializes Maths helper with variable dictionaries.

```
public Maths(Dictionary<string, int> intVars, Dictionary<string, double> realVars)
```

Parameters

`intVars` Dictionary<string, int>

Integer variables.

`realVars` Dictionary<string, double>

Real variables.

Fields

intVars

```
private Dictionary<string, int> intVars
```

Field Value

Dictionary<string, int>

realVars

```
private Dictionary<string, double> realVars
```

Field Value

Dictionary<string, double>

Methods

Evaluate(string)

Evaluates simple expression (+, -, *, /).

```
private double Evaluate(string expr)
```

Parameters

`expr` string

Returns

double

EvaluateInt(string)

Evaluates expression and returns integer result.

```
public int EvaluateInt(string expr)
```

Parameters

`expr` string

Returns

int

EvaluateReal(string)

Evaluates expression and returns real result.

```
public double EvaluateReal(string expr)
```

Parameters

`expr` string

Returns

double

GetValue(string)

Gets value from variables or parses number.

```
private double GetValue(string val)
```

Parameters

`val` string

Returns

double

Class Method

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Simple interpreter that supports integer methods in BOOSE. Methods do not return values explicitly – instead, the result is stored in a variable with the same name as the method.

```
public class Method : IInterpreter
```

Inheritance

object ← Method

Implements

IInterpreter

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Method(AppCanvas)

Constructor - initialises canvas and maths engine.

```
public Method(AppCanvas canvas)
```

Parameters

`canvas` AppCanvas

Fields

addFeatures

```
private AdditionalFeatures addFeatures
```

Field Value

AdditionalFeatures

canvas

```
private AppCanvas canvas
```

Field Value

AppCanvas

intVars

```
private Dictionary<string, int> intVars
```

Field Value

Dictionary<string, int>

maths

```
private Maths maths
```

Field Value

Maths

methods

```
private Dictionary<string, string> methods
```

Field Value

Dictionary<string, string>

parameters

```
private Dictionary<string, string[]> parameters
```

Field Value

Dictionary<string, string[]>

Methods

ExecuteBlock(string)

Executes multiple lines of code (used for method body).

```
private void ExecuteBlock(string block)
```

Parameters

`block` string

ExecuteLine(string)

Executes a single BOOSE command.

```
private void ExecuteLine(string line)
```

Parameters

`line` string

Run(string)

Runs the BOOSE program line by line.

```
public void Run(string program)
```

Parameters

`program` string

Class MoveTo

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

BOOSE command that moves the drawing cursor to a specified position on the canvas.

```
internal class MoveTo : CommandTwoParameters, ICommand
```

Inheritance

object ← Command ← CanvasCommand ← CommandOneParameter ← CommandTwoParameters ← MoveTo

Implements

ICommand

Inherited Members

CommandTwoParameters.param2 , CommandTwoParameters.param2unprocessed , CommandTwoParameters.CheckParameters(string[]) , CommandOneParameter.param1 , CommandOneParameter.param1unprocessed , CanvasCommand.yPos , CanvasCommand.xPos , CanvasCommand.canvas , CanvasCommand.Canvas , Command.IsDouble , Command.program , Command.parameterList , Command.parameters , Command.paramsint , Command.Set(StoredProgram, string) , Command.Compile() , Command.ProcessParameters(string) , Command.ToString() , Command.Program , Command.Name , Command.ParameterList , Command.Parameters , Command.Paramsint , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

MoveTo()

Default constructor used by the command factory.

```
public MoveTo()
```

MoveTo(Canvas, int, int)

Initializes a new instance of the MoveTo command.

```
public MoveTo(Canvas c, int x, int y)
```

Parameters

`c` Canvas

Canvas used for drawing.

`x` int

Target X coordinate.

`y` int

Target Y coordinate.

Fields

x

```
private int x
```

Field Value

int

y

```
private int y
```

Field Value

int

Methods

Execute()

Executes the MoveTo command and moves the drawing cursor.

```
public override void Execute()
```

Exceptions

CanvasException

Thrown when parameters are invalid.

Class Program

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

```
internal static class Program
```

Inheritance

object ← Program

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Main()

The main entry point for the application.

```
[STAThread]
private static void Main()
```

Class Real

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Real variable support for BOOSE interpreter. Allows declaration, assignment and usage of variables in BOOSE commands.

```
public class Real : IInterpreter
```

Inheritance

object ← Real

Implements

IInterpreter

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Real(AppCanvas)

Initializes the interpreter with a drawing canvas.

```
public Real(AppCanvas canvas)
```

Parameters

`canvas` AppCanvas

Canvas used for drawing operations.

Fields

canvas

```
private AppCanvas canvas
```

Field Value

AppCanvas

maths

```
private Maths maths
```

Field Value

Maths

vars

```
private Dictionary<string, double> vars
```

Field Value

Dictionary <string, double>

Methods

Run(string)

Executes a BOOSE program with real variable support.

```
public void Run(string program)
```

Parameters

`program` string

Program text entered by the user.

Class Rectangle

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

BOOSE command that draws a rectangle on the canvas.

```
internal class Rectangle : CommandTwoParameters, ICommand
```

Inheritance

object ← Command ← CanvasCommand ← CommandOneParameter ← CommandTwoParameters ← Rectangle

Implements

ICommand

Inherited Members

CommandTwoParameters.param2 , CommandTwoParameters.param2unprocessed , CommandTwoParameters.CheckParameters(string[]) , CommandOneParameter.param1 , CommandOneParameter.param1unprocessed , CanvasCommand.yPos , CanvasCommand.xPos , CanvasCommand.canvas , CanvasCommand.Canvas , Command.IsDouble , Command.program , Command.parameterList , Command.parameters , Command.paramsint , Command.Set(StoredProgram, string) , Command.Compile() , Command.ProcessParameters(string) , Command.ToString() , Command.Program , Command.Name , Command.ParameterList , Command.Parameters , Command.Paramsint , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

Rectangle()

Default constructor used by the command factory.

```
public Rectangle()
```

Rectangle(Canvas, int, int)

Initializes a new instance of the Rectangle command.

```
public Rectangle(Canvas c, int width, int height)
```

Parameters

`c` Canvas

Canvas used for drawing.

`width` int

Width of the rectangle.

`height` int

Height of the rectangle.

Fields

height

```
private int height
```

Field Value

int

width

```
private int width
```

Field Value

int

Methods

Execute()

Executes the rectangle command and draws the rectangle.

```
public override void Execute()
```

Exceptions

CanvasException

Thrown when parameters are invalid.

Class Tri

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

BOOSE command that draws a triangle on the canvas.

```
internal class Tri : CommandTwoParameters, ICommand
```

Inheritance

object ← Command ← CanvasCommand ← CommandOneParameter ← CommandTwoParameters ← Tri

Implements

ICommand

Inherited Members

CommandTwoParameters.param2 , CommandTwoParameters.param2unprocessed , CommandTwoParameters.CheckParameters(string[]) , CommandOneParameter.param1 , CommandOneParameter.param1unprocessed , CanvasCommand.yPos , CanvasCommand.xPos , CanvasCommand.canvas , CanvasCommand.Canvas , Command.IsDouble , Command.program , Command.parameterList , Command.parameters , Command.paramsint , Command.Set(StoredProgram, string) , Command.Compile() , Command.ProcessParameters(string) , Command.ToString() , Command.Program , Command.Name , Command.ParameterList , Command.Parameters , Command.Paramsint , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

Tri()

```
public Tri()
```

Tri(Canvas, int, int)

```
public Tri(Canvas c, int width, int height)
```

Parameters

`c` Canvas

`width` int

`height` int

Fields

height

```
private int height
```

Field Value

int

width

```
private int width
```

Field Value

int

Methods

Execute()

```
public override void Execute()
```

Namespace BOOSEappTests

Classes

AdditionalFeaturesTests

Unit tests for custom BOOSE additional features. Covers random values, named colours, and filled shapes.

CanvasCommandTests

Unit tests for BOOSE canvas commands and interpreter execution.

ControlFlowTests

Unit tests for BOOSE control flow features. Covers if, else, while, and for support.

MethodNotFoundTests

Unit tests for invalid method calls.

MethodsTests

Unit tests for BOOSE method features. Covers method definition, parameter passing, and method calls.

VariablesTests

Unit tests for BOOSE variable features. Covers int, real, and array support.

WrongParameterTests

Unit tests for invalid method parameter usage.

Class AdditionalFeaturesTests

Namespace: BOOSEappTests

Assembly: BOOSEappTests.dll

Unit tests for custom BOOSE additional features. Covers random values, named colours, and filled shapes.

```
[TestClass]
public class AdditionalFeaturesTests
```

Inheritance

object ← AdditionalFeaturesTests

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

AdditionalFeatures_RandomColourFill_ProgramRunsSuccessfully()

Tests that additional features such as random, colour, and fill execute successfully without throwing an exception.

```
[TestMethod]
public void AdditionalFeatures_RandomColourFill_ProgramRunsSuccessfully()
```

Class CanvasCommandTests

Namespace: BOOSEappTests

Assembly: BOOSEappTests.dll

Unit tests for BOOSE canvas commands and interpreter execution.

```
[TestClass]
public class CanvasCommandTests
```

Inheritance

object ← CanvasCommandTests

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

_canvas

```
private AppCanvas _canvas
```

Field Value

AppCanvas

_factory

```
private AppCommandFactory _factory
```

Field Value

AppCommandFactory

_parser

```
private IParser _parser
```

Field Value

IParser

_program

```
private StoredProgram _program
```

Field Value

StoredProgram

Methods

DrawToCommand_UpdatesPenPositionVariables()

Tests that the DrawTo command correctly updates the pen position variables after drawing a line.

```
[TestMethod]
public void DrawToCommand_UpdatesPenPositionVariables()
```

MoveToCommand_UpdatesPenPositionVariables()

Tests that the MoveTo command correctly updates the pen position variables (Xpos and Ypos).

```
[TestMethod]
public void MoveToCommand_UpdatesPenPositionVariables()
```

MultilineProgram_ExecutesAndLeavesPenAtExpectedPosition()

Tests execution of a multiline BOOSE program and verifies that the final pen position is correct.

```
[TestMethod]
public void MultilineProgram_ExecutesAndLeavesPenAtExpectedPosition()
```

Setup()

Initializes test environment before each test case. Sets up canvas, factory, stored program and parser.

```
[TestInitialize]
public void Setup()
```

Class ControlFlowTests

Namespace: BOOSEappTests

Assembly: BOOSEappTests.dll

Unit tests for BOOSE control flow features. Covers if, else, while, and for support.

```
[TestClass]
public class ControlFlowTests
```

Inheritance

object ← ControlFlowTests

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ControlFlow_IfElseWhileFor_ProgramRunsSuccessfully()

Tests support for if/else statements, while loops, and for loops. The test passes if the program executes without throwing an exception.

```
[TestMethod]
public void ControlFlow_IfElseWhileFor_ProgramRunsSuccessfully()
```

Class MethodNotFoundTests

Namespace: BOOSEappTests

Assembly: BOOSEappTests.dll

Unit tests for invalid method calls.

```
[TestClass]
public class MethodNotFoundTests
```

Inheritance

object ← MethodNotFoundTests

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Methods_MethodNotFound_ThrowsException()

Tests that an exception is thrown when a method is called but has not been defined.

```
[TestMethod]
[ExpectedException(typeof(Exception))]
public void Methods_MethodNotFound_ThrowsException()
```

Class MethodsTests

Namespace: BOOSEappTests

Assembly: BOOSEappTests.dll

Unit tests for BOOSE method features. Covers method definition, parameter passing, and method calls.

```
[TestClass]
public class MethodsTests
```

Inheritance

object ← MethodsTests

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Methods_CallAndExecution_ProgramRunsSuccessfully()

Tests support for method definition, method calls, and parameter passing. The test passes if the program executes without throwing an exception.

```
[TestMethod]
public void Methods_CallAndExecution_ProgramRunsSuccessfully()
```

Class VariablesTests

Namespace: BOOSEappTests

Assembly: BOOSEappTests.dll

Unit tests for BOOSE variable features. Covers int, real, and array support.

```
[TestClass]
public class VariablesTests
```

Inheritance

object ← VariablesTests

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Variables_IntRealArray_ProgramRunsSuccessfully()

Tests support for integer variables, real variables, and arrays. The test passes if the program executes without throwing an exception.

```
[TestMethod]
public void Variables_IntRealArray_ProgramRunsSuccessfully()
```

Class WrongParameterTests

Namespace: BOOSEappTests

Assembly: BOOSEappTests.dll

Unit tests for invalid method parameter usage.

```
[TestClass]
public class WrongParameterTests
```

Inheritance

object ← WrongParameterTests

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Methods_WrongParameterCount_ThrowsException()

Tests that an exception is thrown when a method is called with the wrong number of parameters.

```
[TestMethod]
[ExpectedException(typeof(Exception))]
public void Methods_WrongParameterCount_ThrowsException()
```